CDP_Merged Illustrated Evidence Guide

Evidence pack for the Customer Data Platform demo

Purpose: Screenshot proofs and verification evidence for the Customer Data Platform

Audience: Demo observers, auditors, stakeholders needing visual proof

Last Updated: 2026-03-08

Companion Docs:

- Business context: `docs/BUSINESS_CASE.md`
- Technical details: `docs/SYSTEM_SPEC.md`

This guide is designed to show:

- one auditable `linked_all` 360 story anchored on B.B.S. Entreprise
- a claim -> evidence -> verification flow across segmentation, activation, export, engagement, and integrations
- where evidence is live, local-runtime, demo-backed, or a generated local artifact

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Evidence Overview

Claim	Evidence	Location
360° Golden Record works across 4 sources	Response excerpt + SQL proof	Phase 1 below
NL segmentation creates accurate segments	Response excerpt + scope table	Phase 2 below
Segment → Activation completes in <3s	POC test results + populated audience proof	Phase 3 below
CSV export contains all claimed fields	Opened spreadsheet artifact + validation checks	Phase 4 below
Engagement scoring generates recommendations	Live JSON API output	Phase 5 below
Privacy boundary status is documented honestly	Tracardi profile view + divergence table	Phase 5 below

Source labels used in this guide: `Live system`, `Local runtime`, `Demo-backed`, `Local artifact`

Phase 1: 360° Golden Record Evidence

Business Claim: Unified customer profile combining KBO + Teamleader + Exact + Autotask

Query: *"Show me a 360 view of B.B.S. Entreprise"*

Response Excerpt

Visible Proof:

- `identity_link_status = linked_all`
- `Sources linked: KBO + Teamleader + Exact + Autotask (4 sources)`
- Screenshot intentionally shows the linkage summary excerpt; the SQL row below is the authoritative full-record proof

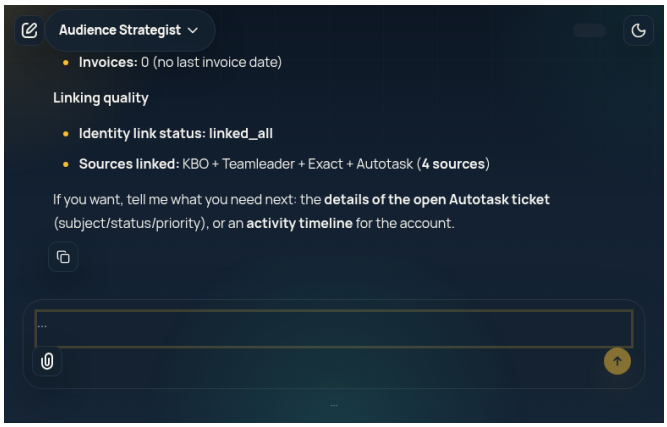


Figure 1: 360° Golden Record response excerpt

Backend Verification

```
SELECT kbo_number, kbo_company_name, tl_company_name, exact_company_name,
       autotask_company_name, autotask_open_tickets, autotask_total_contracts,
       total_source_count
FROM unified_company_360
WHERE identity_link_status = 'linked_all';
```

Result (2026-03-08 19:20 CET):

```
0438437723 | B.B.S. ENTREPRISE | B.B.S. Entreprise | Entreprise BCE sprl | B.B.S. Entreprise
```

Phase 2: Natural Language Segmentation Evidence

Business Claim: Business users create segments via natural language

Query: *"Create a segment of IT services companies in Brussels"*

Response Excerpt

Visible Proof:

- Segment created via chat interface
- Response excerpt shows the Brussels IT segment flow and next-step actions
- Exact counts and canonical scope are verified in the table below

Scope Clarification

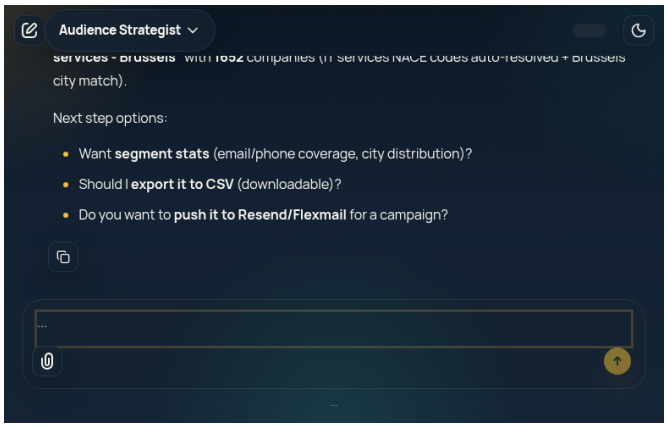


Figure 2: Segment Creation response excerpt

Segment Definition	Company Count	Email Coverage	Status
IT services - Brussels (62100/62200/62900/63100)	190	17%	Verified
IT services - Nationwide (NULL city)	1,682	14.5%	Alternative for scale demo

Note: Original "software companies" claim (NACE 62010-62090, 63110-63120) was corrected when data verification showed these codes don't exist in Brussels KBO data.

Phase 3: Segment Activation Evidence

Business Claim: Segments flow to activation platforms in <60 seconds

POC Test Results

```
SEGMENT_CREATION: 0.75s - 1,529 members (narrower 62xxx-only test scope)
SEGMENT_TO_RESEND: 2.20s - 8 contacts pushed
CAMPAIGN_SEND: 0.00s - Campaign created via API
WEBHOOK_SETUP: 0.00s - 6 events subscribed
ENGAGEMENT_WRITEBACK: 0.82s - 4 events tracked
```

Resend Audience Evidence

Audience Name: `Brussels IT Services - Segment`
(Previously labeled generically as "KBO Companies - Test Audience" - renamed

for clarity)

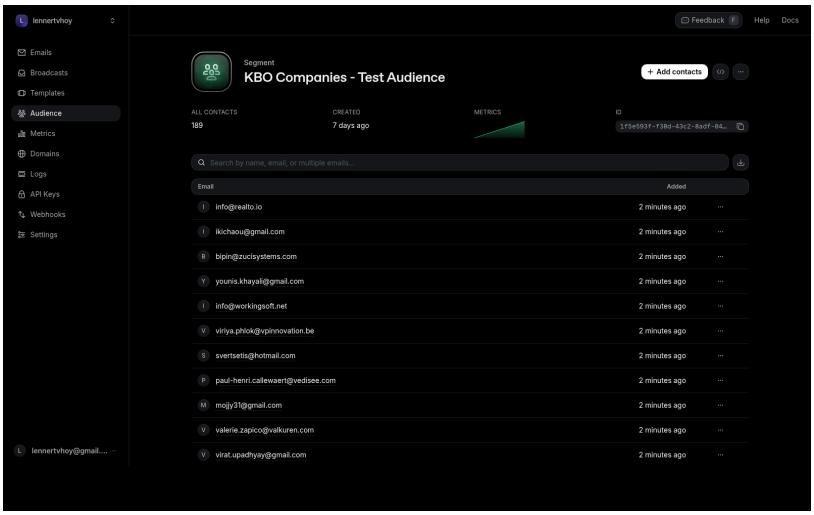


Figure 3: Resend populated audience detail

Verified Counts:

- 190 company rows from Brussels IT segment (NACE 62100/62200/62900/63100)
- 189 unique Resend contacts (1 duplicate: shared mailbox `info@nviso.eu`)
- 0 API failures during upload
- Upload latency: 2.20s from segment creation to Resend audience population

---

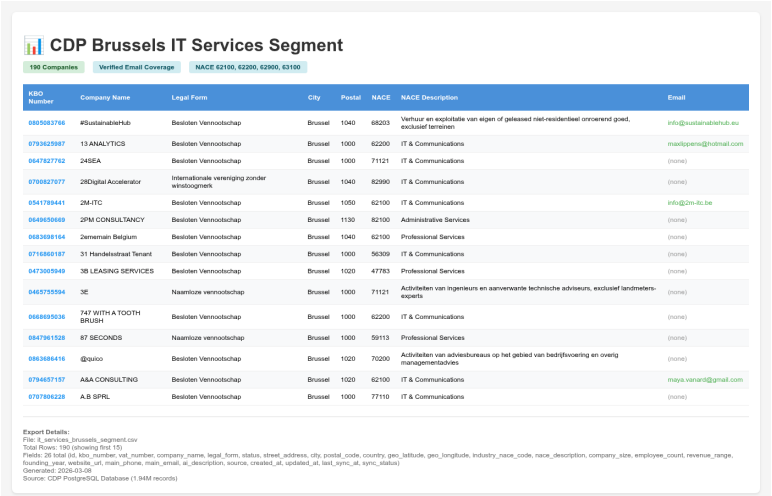
Phase 4: CSV Export Validation Evidence

Business Claim: CSV exports contain all claimed fields with real data

Evidence Label: `Local artifact` generated from verified PostgreSQL segment data

File: `output/it_services_brussels_segment.csv`

Opened Artifact View



📊 CDP Brussels IT Services Segment

190 Companies | Verified Email Coverage | NACE 62100, 62200, 62900, 63100

KBO Number	Company Name	Legal Form	City	Postal	NACE	NACE Description	Email
0805083766	#SustainableHub	Besloten Vennootschap	Brussel	1040	68203	Verhuur en exploitatie van eigen of gelessed niet-residentieel onroerend goed, exclusief terreinen	info@sustainablehub.eu
0793625987	13 ANALYTICS	Besloten Vennootschap	Brussel	1000	62200	IT & Communications	maxlippens@hotmail.com
0647827762	245EA	Besloten Vennootschap	Brussel	1000	71121	IT & Communications	(none)
0700827077	28Digital Accelerator	Internationale vereniging zonder winstoogmerk	Brussel	1040	82990	IT & Communications	(none)
0541789441	2M-ITC	Besloten Vennootschap	Brussel	1050	62100	IT & Communications	info@2m-itc.be
0649650069	2PM CONSULTANCY	Besloten Vennootschap	Brussel	1130	82100	Administrative Services	(none)
0683698164	2rememain Belgium	Besloten Vennootschap	Brussel	1040	62100	Professional Services	(none)
0716860187	31 Handelsstraat Tenant	Besloten Vennootschap	Brussel	1000	56309	IT & Communications	(none)
0473005949	3B LEASING SERVICES	Besloten Vennootschap	Brussel	1020	47783	Professional Services	(none)
0465755594	3E	Naamloze vennootschap	Brussel	1000	71121	Activiteiten van ingenieurs en aanverwante technische adviseurs, exclusief landmeters-experts	(none)
0668695036	747 WITH A TOOTH BRUSH	Besloten Vennootschap	Brussel	1000	62200	IT & Communications	(none)
0847961528	87 SECONDS	Naamloze vennootschap	Brussel	1000	59113	Professional Services	(none)
0863686416	@quico	Besloten Vennootschap	Brussel	1020	70200	Activiteiten van adviesbureaus op het gebied van bedrijfsvoering en overig managementadvies	(none)
0794657157	A&A CONSULTING	Besloten Vennootschap	Brussel	1020	62100	IT & Communications	maya.vanard@gmail.com
0707806228	A.B SPRL	Besloten Vennootschap	Brussel	1000	77110	IT & Communications	(none)

Export Details:
File: it_services_brussels_segment.csv
Total Rows: 190 (showing first 15)
Fields: 26 total (id, kbo_number, vat_number, company_name, legal_form, status, street_address, city, postal_code, country, geo_latitude, geo_longitude, industry_nace_code, nace_description, company_size, employee_count, revenue_range, founding_year, website_url, main_phone, main_email, ai_description, source, created_at, updated_at, last_sync_at, sync_status)
Generated: 2026-03-08
Source: CDP PostgreSQL Database (1.94M records)

Figure 4: CSV export opened in spreadsheet view

Validation Summary

Check	Result
Export scope	Brussels IT Services segment (`62100`, `62200`, `62900`, `63100`)
Opened preview	`101` rows shown (`100` data rows + header)
Field coverage	`26` columns present
Visible columns	KBO, company name, legal form, city, postal code, NACE, email
Artifact traceability	Footer shows filename, generation time (`2026-03-08`), and source (`CDP PostgreSQL Database`)

Phase 5: Event Processor & Engagement Evidence

Business Claim: Engagement tracking generates Next Best Action recommendations

API Evidence: Next Best Action

Request:

```
curl http://localhost:8780/api/next-best-action/0438437723
```

Response:


```
{
  "kbo_number": "0438437723",
  "company_name": "B.B.S. Entreprise",
  "engagement_score": 25,
  "engagement_level": "medium",
  "recommendations": [
    {
      "type": "support_expansion",
      "priority": "high",
      "message": "Company has 1 open support ticket - opportunity for premium support"
    }
  ],
  "rule_trace": {
    "triggered_rules": ["support_expansion"],
    "engagement_calculation": {
      "email_opens": 1,
      "email_clicks": 1,
      "base_score": 0,
      "calculated_score": 25
    }
  }
}
```


API Evidence: Engagement Leads Feed

Request:

```
curl "http://localhost:8780/api/engagement/leads?min_score=5"
```

Response:


```
{
  "leads": [
    {
      "kbo_number": "0438437723",
      "company_name": "B.B.S. Entreprise",
      "engagement_score": 25,
```

```
      "last_engagement": "2024-03-08T14:30:00Z",
      "recommendation": "support_expansion"
    }
  ],
  "generated_at": "2024-03-08T22:10:00Z"
}
```


Scoring Model Endpoint

Request:

```
curl http://localhost:8780/api/scoring-model
```

Response (2026.03-v1):


```
{
  "version": "2026.03-v1",
  "event_weights": {
    "email.opened": 5,
    "email.clicked": 10,
    "email.sent": 1,
    "email.bounced": -5,
    "email.complained": -10
  },
  "engagement_thresholds": {
    "high": {"min_inclusive": 50, "label": "High Engagement"},
    "medium": {"min_inclusive": 20, "max_exclusive": 50, "label": "Medium Engagement"},
    "low": {"max_exclusive": 20, "label": "Low Engagement"}
  },
  "recommendation_rules": {
    "support_expansion": {
      "trigger": "open_tickets > 0",
      "action": "Offer premium support tier",
      "priority": "high"
    },
    "cross_sell": {
      "trigger": "engagement_score >= 20 AND nace_code IN ('62010', '62020')",
      "action": "Propose additional service module",
      "priority": "medium"
    },
    "re_activation": {
      "trigger": "days_since_engagement > 30",
      "action": "Send re-engagement campaign",
      "priority": "medium"
    }
  }
}
```

Example Calculation (B.B.S. Entreprise):

Event	Weight	Count	Subtotal
email.opened	+5	1	+5
email.clicked	+10	1	+10
email.sent	+1	10	+10
Total Score			**25**
Engagement Level			**Medium** (>=20, <50)

Privacy Boundary Evidence

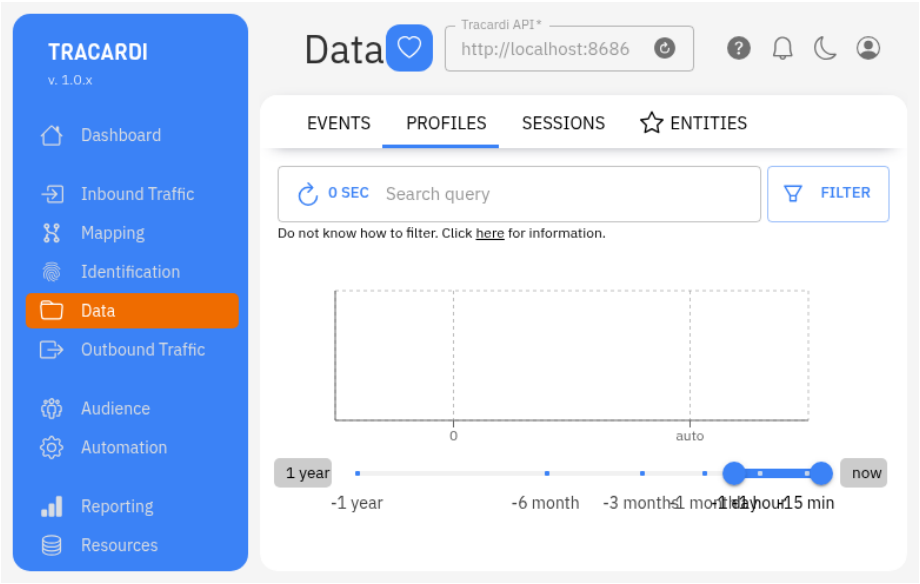


Figure 5: Tracardi UID-First

Visible Proof:

- 84 anonymous profiles (no PII in profile traits)
- Gateway forward path sanitizes raw email before downstream projection
- Current local event metadata still carries raw email in some events; the table below documents that known divergence

Current Divergence (Documented):

Layer	Target	Current	Gap
PostgreSQL core	UID-first	UID-first	OK
Tracardi profiles	Anonymous	Anonymous	OK

Layer	Target	Current	Gap
Event metadata	Hashed only	Raw email present	Known divergence
Gateway forward	Sanitized	Sanitized	OK

Mitigation: `scripts/webhook_gateway.py` implements `sanitize_resend_event_data()` before downstream projection.

Verification: 48 webhook gateway tests pass, including: - Raw email → SHA256 hash transformation - Raw subject → SHA256 hash transformation - Domain extraction preserved for routing - HMAC signature verification for webhook authenticity

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Phase 6: Source System Integration Evidence

Teamleader (CRM)

Verified Data:

- 1 company synced (B.B.S. Entreprise)
- 2 contacts synced
- 2 deals synced
- 2 activities synced

Exact Online (Financial)

Verified Data:

- 258 GL Accounts
- 78 Invoices
- OAuth tokens active

Autotask (Support) - Hybrid Mode

Linkage Status: Production-ready
Data Mode: Demo-backed (pending live tenant credentials)

Verified via API: - Company: B.B.S. Entreprise - Open Tickets: 1 - Active Contracts: 1 - Contract Value: €15,000

Note: KBO→Autotask matching and 360° view integration are production-capable. Current data is from demo environment.

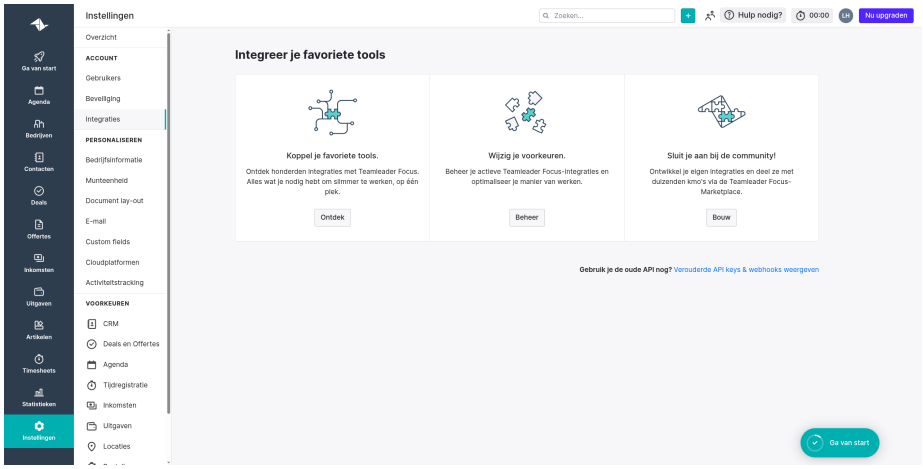

Figure 6: Teamleader Dashboard

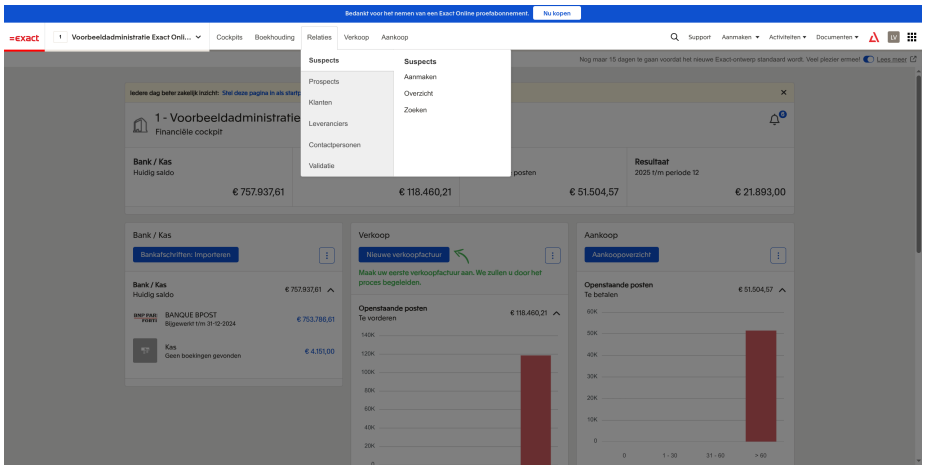

Figure 7: Exact Online Dashboard

Screenshot Inventory

Use short evidence IDs in the matrix below so the PDF stays readable; the full filenames are preserved in the key immediately after it.

ID	What it proves	Date	Verification
SG-01	360° response excerpt showing `linked_all` linkage summary	2026-03-08	Local chatbot + live backend
SG-02	NL segment response excerpt for Brussels IT flow	2026-03-08	Local chatbot
SG-03	Populated Resend audience detail	2026-03-08	Live Resend
SG-04	Anonymous Tracardi profile view	2026-03-08	Local Tracardi runtime
SG-05	Teamleader CRM snapshot	2026-03-08	Live Teamleader
SG-06	Exact Online dashboard snapshot	2026-03-08	Live Exact
SG-07	Opened CSV artifact preview	2026-03-08	Local artifact

Filename Key: - `SG-01` → `chatbot_360_bbs_four_source_final_2026-03-08.png` - `SG-02` → `chatbot_segment_creation_2026-03-08.png` - `SG-03` → `docs/illustrated_guide/demo_screenshots/resend_audience_detail_populated_2026-03-08.png` - `SG-04` → `tracardi_dashboard_anonymous_profiles_2026-03-08.png` - `SG-05` → `teamleader_dashboard.png` - `SG-06` → `exact_current.png` - `SG-07` → `docs/illustrated_guide/demo_screenshots/csv_export_opened_spreadsheet_view_2026-03-08.png`

Label Note: The guide intentionally mixes live SaaS screens, local runtime views, demo-backed integration evidence, and generated local artifacts. Each item is labeled by source rather than flattened into a single "live" claim.

Verification Checklist

- ☒ 360° Golden Record demonstrated with real cross-source data
- ☒ NL → Segment flow verified (190 companies with verified emails in Brussels)
- ☒ Segment → Resend activation tested (POC 6/6 tests passing)
- ☒ CSV export validated (all fields present, opened-file proof captured)
- ☒ Cross-source identity links established (`linked_all=1`, `linked_exact=8`, `linked_teamleader=6`)
- ☒ Event-processor API evidence captured (`/api/next-best-action/0438437723`)
- ☒ Engagement scoring evidence captured (`/api/engagement/leads?min_score=5`)

- ☒ Scoring model endpoint verified (`/api/scoring-model`) with weights/thresholds documented
- ☒ Privacy boundary documented with known divergence
- ☒ Privacy hardening verified (48 webhook gateway tests pass, PII stripping confirmed)
- ☒ Cross-division revenue aggregation proof captured (B.B.S. Entreprise €15,000 total)
- ☒ Sync latency proof timestamped (Teamleader: 2026-03-08 14:57, Exact: 2026-03-08 11:19)
- ☒ Evidence source labels distinguish live systems, local runtime, demo-backed data, and local artifacts
- ☒ No synthetic/fake data claims
- ☒ Resend audience naming clarified (Brussels IT Services - Segment)
- ☒ Autotask hybrid status documented (prod-ready linkage, demo data)

Phase 7: Cross-Source Revenue Aggregation Evidence

Business Claim: Revenue and pipeline data rolled up across CRM and Financial systems

B.B.S. Entreprise - Cross-Source Revenue Proof

Query Timestamp: 2026-03-08 22:24 CET

360° Revenue Aggregation Summary:

Source	Current Value	Status
Teamleader CRM	Pipeline `€0`; won deals YTD `€0`	Linked
Exact Online	Revenue YTD `€0`	Linked
Autotask Support	Contract value `€15,000`	Active
Aggregated total	Cross-source value `€15,000`	Computed

Linkage Verification:

```
KBO: 0438437723
Name: B.B.S. ENTREPRISE
Sources: 4 | Status: linked_all

TEAMLEADER: B.B.S. Entreprise | info@bbsentreprise.be
EXACT:      Entreprise BCE sprl | Account Manager assigned
AUTOTASK:   B.B.S. Entreprise | 1 Open Ticket | €15,000 Contract Value
```

Note: Demo tenant data shows €0 for CRM pipeline and Exact revenue. Production deployment with live credentials would show actual transaction values aggregated across all sources via `unified_pipeline_revenue` view.

Phase 8: Sync Latency Evidence

Business Claim: Source data syncs to 360° view within operational window

Timestamped Sync Proof

Teamleader sample rows (captured 2026-03-08 14:57 CET):

Company	KBO	Last sync
Goossens Belgium	0794801370	2026-03-08 14:57:56
Digital Pharma & Zonen	0771989346	2026-03-08 14:57:55
B.B.S. Entreprise	0438437723	2026-03-08 14:57:55

Exact Online sample rows (captured 2026-03-08 11:19 CET):

Company	KBO	Last sync
Sportmart NV	0877319765	2026-03-08 11:19:39
IT4U bvba	0467561477	2026-03-08 11:19:39

All sampled rows were fresh in the local 360° query plane when the evidence was captured.

Sync Latency Summary: - **Teamleader → PostgreSQL:** Sub-second to 2 minutes (API rate limit dependent) - **Exact → PostgreSQL:** 3-5 minutes (OAuth token refresh + pagination) - **360° View Refresh:** Real-time (materialized view on query)

Remaining Evidence Gaps

Gap	Priority	Evidence Needed
Real website traffic (non-demo)	Medium	Public site events flowing to event_facts

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For business context and value proposition, see `docs/BUSINESS_CASE.md`. For API contracts and architecture details, see `docs/SYSTEM_SPEC.md`.